

SPATIAL FEATUIRES OF URBAN ENVIRONMENT PROMOTING HEALTH

Mara Lúcia Marques², Pontifícia Universidade Católica (PUC), Campinas, São Paulo, Brasil.

Dalton Müller Pessoa Filho⁶, Universidade Estadual Paulista (UNESP), Bauru, São Paulo, Brasil.

E-mails dos autores:

mara.marques@puc-campinas.edu.br

dalton.pessoa-filho@unesp.br

Introduction

The incentive to practice physical activities in public environments is a trend in health policy strategies, aiming at reducing the propensity for chronic noncommunicable diseases and promoting healthy habits by increasing physical activity (PA) among the population (BLAIR, 2007).

For WHO (2016) one of the challenges of this type of approach is to detect indicators that combine environmental features and health. Therefore, the objective of the present study was to develop an urban index to identify the city's propensity to practice PA, intending to relate it to obesity anthropometric indicators and exercise structure parameters, as well as to contextualize the life habits of the analyzed population.

Methods

Geographic contextualization

Determine social and demographical features. For the present study, the municipality of Jataí is located in the southwest region of the state of Goiás, Brazil, has an area of 7,174.225 km², and an estimated population in 2017 of 98,128 people

(IBGE, 2017). Its relief is flat and mildly undulating, with an altimetric variation between 500 and 1,100 m. Depicted localization area Figure.

Data sampling

Determine the local. For the present study, field data collection was carried out in urban public environments that showed the highest attendance rates of PA practitioners and with suitable structures such as walking/running tracks and outdoor gyms.

To identify the spatial urban health index (SUHI), data on demographic, environmental, and urban structure were collected. For the present study were consider number of residents (IBGE, 2011), parameters regarding road connectivity and land use and vegetation cover (SIEG, 2018), and the urban area slope parameter generated by the Shuttle Radar Topography Mission (MDE – SRTM), with a spatial resolution of 30 m (EMBRAPA, 1979). Table 1 describes the parameters and indicators used in the definition of the SUHI (Equation 1), as well as the weighting of these indicators ranging from 1 to 10. Figure 2 depicts all products generated from these data by using Geographical information system (GIS).

$$SUHI = D_{populational} + D_{green\ area} + Slope + D_{roads\ connection} \quad (1)$$

Table 1: Criteria for urban pattern classification.

Parameters	Indicators	Variables	Rate
Number of people per sector	Population density (people/km ²) <i>D_{population}</i>	4 – 300	1
		300 – 1300	1
		1300 – 2500	3
		2500 – 3500	4
		3500 – 4500	5
		4500 – 5500	7
		5500 – 8500	8
		8500 – 14197	10

Urban green space per people	Urban green space density (green m ² /people) <i>D_{green area}</i>	< 12	1
		12 – 20	5
		20 – 36	8
		> 36	10
Relief	Surface slope (%) <i>Slope</i>	0 - 3 (very gently)	10
		3 - 8 (gently)	9
		8 - 20 (moderate)	5
		20 - 45 (steep)	2
		> 45 (very steep)	1
Connectivity roads	Road density (road km/km ²) <i>D_{road connection}</i>	< 5	1
		5 – 15	5
		15 – 25	7
		25 – 35	9
		35 – 45	10

Use GIS. For the presente study, a geographic database was prepared with GIS ArcGis 10.5.1 (ESRI v. 2018) for the composition of the physical and demographic analysis parameters (Table 1) and mapping of the SUHI by using the weighted multi-criteria overlay method (Equation 1), which allowed to obtain a representativeness index (km²), classified by applying the quantile method, into four gradation levels ranging from unfavorable to very favorable to PA (Figure 2 – the last Panel on the botton).

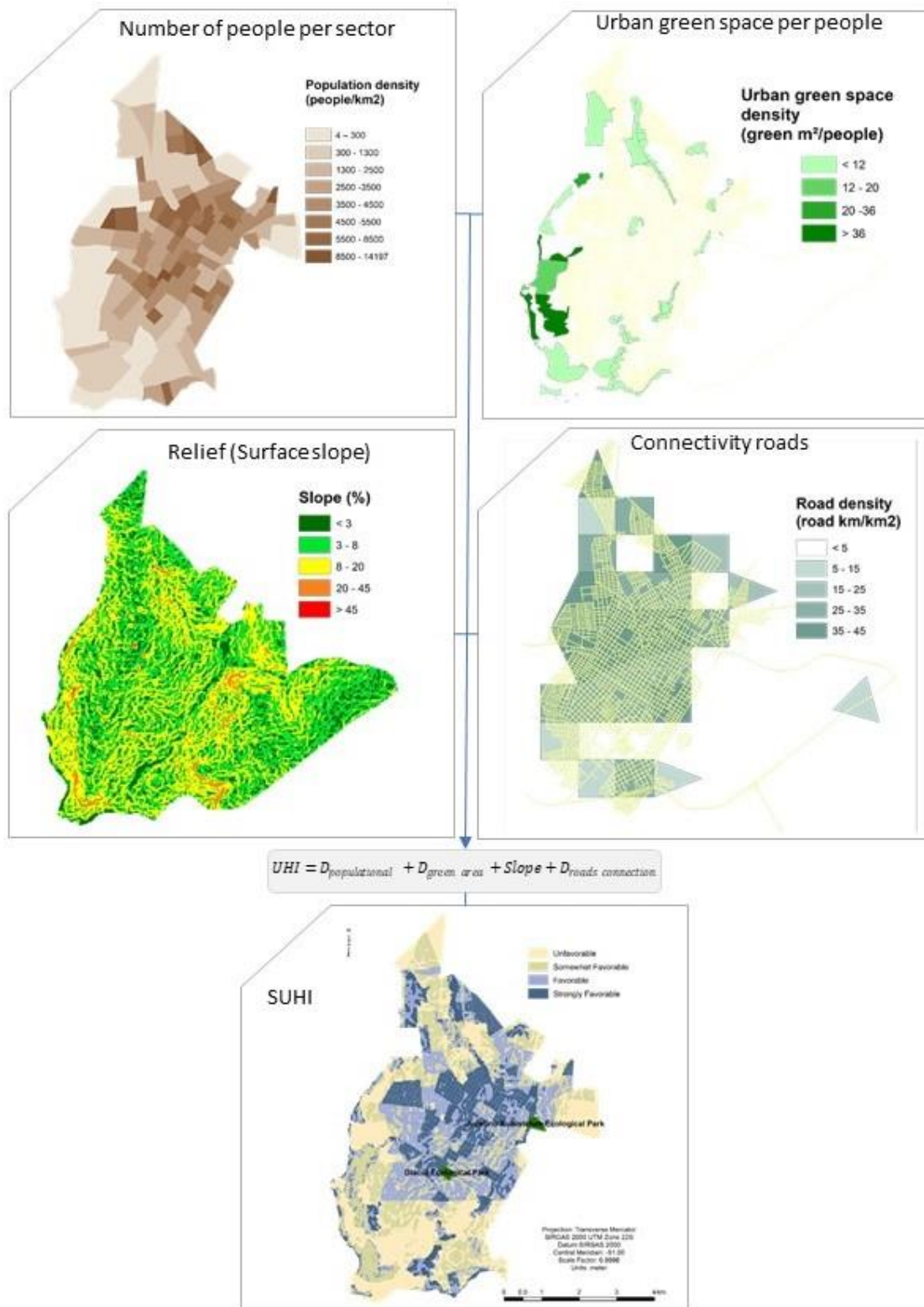


Figure 2: Products of Population density, Urban green space density, Surface slope, Road density and the Classification of urban environment ability to promote physical activity according to spatial and demographical characteristics.

REFERÊNCIAS

BLAIR, Steven N. 61 Physical inactivity: the biggest public health problem of the 21st Century. **J Sci Med Sport**, 10:29-33, 2007.

IBGE, Instituto Brasileiro de Geografia e Estatística. Estimativa de População: População Estimada. IBGE: Instituto Brasileiro de Geografia e Estatística, 2017.

WHO, World Health Organization. Waist Circumference and Waist-Hip Ratio: report of a WHO expert consultation. Geneva: 2008.

EMBRAPA. Serviço nacional de Levantamento e Conservação de Solos. Manual de métodos de análise de solo. Rio de Janeiro, 1979. 80p.

EURO, WHO. Urban green spaces and health: A review of the evidence. Copenhagen: WHO Regional Office for Europe, 2016.

IBGE - Censo Demográfico 2010. Resultados do universo por setores censitários, 2011.

MARTINS, A. P., OLIVEIRA, R. M. (org.) Diagnóstico ambiental das microbacias hidrográficas da área urbana de Jataí – Go. Universidade Federal de Goiás, 2012.

SIEG- Secretaria de Estado de Gestão e Planejamento – SEGPLAN, 2018.